Jack Hollis Farrell

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Department of Physics, University of Colorado Boulder
Center for Theory of Quantum Matter
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Education

Ph.D. Physics, University of Colorado Boulder — 2021–Present

- Advisor: Prof. Andrew Lucas
- GPA 4.00/4.00

Honours B.Sc. Physics, University of Toronto — 2017–2021

- Major in Physics, Minors in Mathematics and English.
- GPA 3.96/4.00

Articles

In Preparation

1. **Jack H. Farrell** and Andrew Lucas. "Endpoint of a ballistic Dyakonov-Shur instability". To appear (2026).
2. **Jack H. Farrell**, Andrew Hicks, and Andrew Lucas. "Electron hydrodynamics with an open Fermi surface". To appear (2026).
3. Elijah G. Lew-Smith, Marvin Qi, Aaron J. Friedman, **Jack H. Farrell**, and Andrew Lucas. "Constant density flocking without long range order,". To appear (2026).

Publications and Preprints

1. **Jack H. Farrell** and Andrew Lucas. "Characterizing electronic scattering rates with transport in multiterminal devices", arXiv: 2605.03030 (2026).
2. Ludwig Holleis, Youngjoon Choi, Canxun Zhang, **Jack H. Farrell**, Gabriel Bargas, Audrey Hsu, Zexing Chen, Ian Sackin, Wenjie Zhou, Yi Guo, Thibault Charpentier, Yifan Jiang, Benjamin A. Foutty, Aidan Keough, Martin E. Huber, Takashi Taniguchi, Kenji Watanabe, Andrew Lucas, and Andrea F. Young. "Cryogenic shock exfoliation for ultrahigh mobility rhombohedral graphite nanoelectronics", arXiv: 2604.21912 (2026).
3. Canxun Zhang, Evgeny Redekop, Hari Stoyanov, **Jack H. Farrell**, Sunghoon Kim, Ludwig Holleis, David Gong, Yongjoon Choi, Takashi Taniguchi, Kenji Watanabe, Martin E. Huber, Ania C. Bleszynski Jayich, Andrew Lucas, and Andrea F. Young. "Imaging electron hydrodynamics in a graphene flat band system", arXiv: 2603.11175 (2026).
4. Johannes Geurs, Tatiana A. Webb, Yinjie Guo, Itai Keren, **Jack H. Farrell**, Jikai Xu, Kenji Watanabe, Takashi Taniguchi, Dmitri N. Basov, James Hone, Andrew Lucas, Abhay Pasupathy, Cory R. Dean. "Supersonic flow and hydraulic jump in an electronic de Laval nozzle", arXiv: 2509.16321 (2025).

5. Xiaoyang Huang, **Jack H. Farrell**, Aaron J. Friedman, Isabella Zane, Paolo Glorioso, and Andrew Lucas. "Generalized time-reversal symmetry and effective theories for nonequilibrium matter", arXiv: 2310.12233 (2023).
6. **Jack H. Farrell**, Xiaoyang Huang, and Andrew Lucas. "Hydrodynamics with helical symmetry", arXiv: 2208.12269 (2022).
7. **Jack H. Farrell**, Nicolas Grisouard, and Thomas Scaffidi. "Terahertz radiation from the Dyakonov-Shur instability of hydrodynamic electrons in a Corbino geometry". Phys. Rev. B 106, 195432 (2022).

Talks

Invited

1. "Computational Fluid Mechanics for Electrons", Centre de Physique Théorique, École Polytechnique (2026) — Oral presentation.
2. "Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter", CTQM Seminar, Center for Theory of Quantum Matter (2024) — Oral presentation.

Contributed

1. "2D Electron transport in complex geometries", Correlated Quantum Systems workshop, Institut Pascal (2026) — Oral presentation.
2. "2D Electron transport through complex geometries", APS Global Physics Summit (2026) — Oral presentation.
3. "Endpoint of a Ballistic Dyakonov-Shur Instability", APS March Meeting (2025) — Oral presentation.
4. "Generalized Time-Reversal Symmetry and Effective Theories for Nonequilibrium Matter", UQM: Ultra Quantum Matter (2024) — Poster presentation.
5. "Constant density flocking without long range order", Quantum, Classical, and Active Fluids workshop, CUNY Graduate Center (2024) — Oral presentation (video).
6. "Effective Theories for Nonequilibrium Matter", APS March Meeting (2024) — Oral presentation.
7. "Hydrodynamics with Helical Symmetry", APS March Meeting (2023) — Oral presentation.

Honours

NSERC Canada Graduate Scholarship (CGS-M), NSERC — 2021

- Declined: offered at U of T, UBC, University of Waterloo / Perimeter Institute.
- Research grant for first-year Ph.D. students.

University of Toronto National Scholarship — 2017–2021

- Full scholarship to U of T, covering tuition and living fees for four years.

NSERC Undergraduate Summer Research Award, University of Toronto — 2020

Hymie and Roslyn Mida Award in Theoretical Physics, University of Toronto — 2020

NSERC Undergraduate Summer Research Award, University of Toronto — 2019

University of Toronto Scholar 2018–2019

Teaching

Instructor of Record, University of Colorado Boulder

- PHYS 1400: Fundamentals of Scientific Inquiry (course webpage) 2025

Guest Lecturer, University of Colorado Boulder

- PHYS 7810: Graduate Special Topics: Hydrodynamics as an Effective Field Theory 2026
- PHYS 5210: Graduate Theoretical Mechanics 2024
- PHYS 4410: Quantum Mechanics II 2023

Teaching Assistant, University of Colorado Boulder

- PHYS 7810: Graduate Special Topics in Physics: Hydrodynamics Spring 2024
- PHYS 3090: Introduction to Quantum Computing Spring 2024
- PHYS 1110: General Physics I Fall 2023
- PHYS 1125: General Physics II for Majors Spring 2023
- PHYS 1110: General Physics I Fall 2022
- PHYS 1110: General Physics I Spring 2022
- PHYS 1110: General Physics I Fall 2020

Teaching Assistant, University of Toronto 2020–2021

- Extra help for programming in `Python` for physics majors

Service

Audio & Visual Coordinator, Boulder Summer School for Condensed Matter and Materials Physics

- Geometry and Topology in Soft Matter 2026
- Dynamics of Strongly Correlated Electrons 2025
- Self-Organizing Matter: From Inanimate to the Animate 2024
- Non-Equilibrium Quantum Dynamics 2023
- Hydrodynamics Across Scales 2022

Mentoring & Supervision

Research mentoring, University of Colorado Boulder

- Julia Guo, Boulder High School 2026–2027
- Zachary Hill, UT Austin REU 2026–2027
- Andrew Hicks, CU Boulder, undergraduate (now Ph.D. student, Yale University) 2022–2024
- Elijah G. Lew-Smith, Brown U. REU (now Ph.D. student, MIT) 2023–2024

Professional Memberships

Member, American Physical Society (APS) 2021–Present